Experimental Skills and Data Analysis

International Physics Olympiad · Worked Solutions

Introductory study guide · 6 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

These worked solutions accompany the original MAIS PAC problems. Check your assumptions and units.

1. Twenty oscillations take 40.0 s. Find the period.

   Solution. T = 40.0/20 = 2.00 s. Timing many cycles reduces the relative effect of start–stop reaction time.

2. A length is (10.0 ± 0.1) cm. What is its relative uncertainty?

   Solution. 0.1/10.0 = 0.01, or 1%. State whether the uncertainty is absolute or relative.

3. For y = ax + b, two points on a fitted line are (1, 3) and (3, 7). Find slope and intercept.

   Solution. a = (7−3)/(3−1) = 2 and b = 3−2×1 = 1. These points are on the fitted line, not necessarily raw data.

4. Every reading has the same unknown zero offset. Will repeating the measurement remove it?

   Solution. No. Repetition reduces random scatter but the shared offset is systematic; calibrate or measure the zero independently.

5. Challenge 1 — A pendulum has $T^2 = (4\pi^2/g)L$. A line fit of T^2 versus L gives slope (4.00 ± 0.08) s^2/m. Estimate g and its uncertainty using first-order propagation.

   Solution. $g = 4\pi^2/\text{slope} = 9.87\ m/s^2$. Since g is inversely proportional to slope, the relative uncertainty is 0.08/4.00 = 0.02. Therefore $\Delta g \approx 0.20\ m/s^2$ and $g = (9.87 \pm 0.20)\ m/s^2$. This assumes the slope uncertainty captures the relevant errors and the model is valid.

6. Challenge 2 — A resistor is measured with a voltmeter of finite resistance connected in parallel with it, and an ammeter measuring the total current supplied to both branches. Does V/I overestimate or underestimate the resistor’s true resistance? Explain.

Solution. The ammeter reads I = V/R + V/Rv. Thus V/I = RRv/(R+Rv), smaller than R for finite positive Rv. A high-resistance voltmeter reduces this loading bias; repeated readings alone cannot remove it.